Maggie Makar

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Education

Massachusetts Institute of Technology 06/2021
Ph.D, Electrical Engineering and Computer Science

Massachusetts Institute of Technology 05/2017
Master of Science, Electrical Engineering and Computer Science

University of Massachusetts, Amherst 02/2013
B.S., Mathematics and Economics, *Summa Cum Laude*

Employment

Presidential Post-doc/Assistant Professor, University of Michigan, Ann Arbor	08/2021 - Present
Consulting research scientist, Microsoft Research, Redmond	07/2021 - 07/2022
Research Intern, Google Brain	06/2020 - 09/2020
Research Intern, Microsoft Research	06/2018 - 09/2018
Research Assistant, Brigham and Women's Hospital	02/2013 - 08/2015
Research Assistant, Political Economy Research Institute	09/2012 - 12/2012
Research Intern, Donahue Institute,	08/2012 - 09/2012
Research Assistant, Center for Intelligent Information Retrieval	02/2012 - 08/2012

Peer-reviewed publications

* denotes equal contribution.

1. **M. Makar** & Alex D'Amour, "Fairness and Robustness in anti-causal prediction", *ICML workshop 2022*
2. J. Zheng & **M. Makar** "Causally motivated multi-shortcut identification and removal", *ICML workshop 2022*
3. S. Tang, **M. Makar**, M. Sjoding, F. Doshi-Velez, J. Wiens, "Leveraging Factored Action Spaces for Efficient Offline Reinforcement Learning in Healthcare", *ICML workshop 2022*
4. **M. Makar**, B. Packer, D. Moldovan, D. Blalock, Y. Halpern, A. D'Amour, "Causally motivated shortcut removal using auxiliary labels". *AIStats 2022; ACIC poster 2022, ICML workshop 2022*
5. **M. Makar**, L. West, D. Hooper, E. Horvitz, E. Shenoy, J. Guttag, "Exploiting structured data for learning contagious diseases under incomplete testing". *ICML 2021*
6. **M. Makar**, F. Johansson, J. Guttag, D. Sontag, "Estimation of Bounds on Potential Outcomes For Decision Making". *ICML 2020*
7. **M. Makar**, A. Swaminathan, E. Kiciman, "A Distillation Approach to Data Efficient Individual Treatment Effect Estimation". *AAAI, 2019; NeuRIPS workshop 2018*

8. E. Mu, **M. Makar**, L. West, J. Guttag, D. Rosenberg, E. Shenoy. "Learning the Influence of Individual Clostridioides difficile Infections". *Open Forum Infectious Diseases, 2019*
9. **M. Makar**, J. Guttag, J. Wiens, "Learning the Probability of Activation in the Presence of Latent Spreaders". *AAAI, 2018 (spotlight presentation); NeuRIPS workshop 2017*
10. J. Oh*, **M. Makar***, et. al, "A generalizable, data-driven approach to predict daily risk of Clostridium difficile infection at two large academic health centers". *Infection Control and Hospital Epidemiology, 2018*
11. K. Joynt, **M. Makar**, A. Gawande, D. Cutler, Z. Obermeyer, "The Most Expensive Patients in the Hospital", *Circulation, 2018*
12. **M. Makar***, J. Antonelli*, Q. Di, J. Schwartz, D. Cutler, F. Dominici, "Estimating the Causal Effect of Fine Particulate Matter Levels on Death and Hospitalization: Are Levels Below the Safety Standards Harmful?". *Epidemiology, 2017; JSM poster 2016*
13. **M. Makar***, J. Oh*, et. al, "A data-driven approach to predict daily risk of Clostridium difficile infection at two large academic health centers" *Open Forum Infectious Diseases, 2017*
14. Z. Obermeyer, **M. Makar**, S. Abujaber, F. Dominici, S. Block, D. Cutler, "Association between the Medicare hospice benefit and health care utilization and costs for patients with poor-prognosis cancer". *Journal of the American Medical Association, 2014*
15. **M. Makar**, M. Ghassemi, D. Cutler, Z. Obermeyer, "Short-term mortality prediction for elderly patients using Medicare claims data". *International Journal of Machine Learning and Computing, 2015*
16. B. Powers, **M. Makar**, D. Cutler, Z. Obermeyer, "Cost Savings Associated with Expanded Hospice Use Among Medicare Beneficiaries with Poor Prognosis Cancer". *Journal of Palliative Medicine, 2015*
17. Z. Obermeyer, S. Abujaber, **M. Makar**, S. Stoll, S. Kayden, L. Wallis, T. Reynolds, "Emergency care delivery in 59 low- and middle-income countries: Systematic review". *WHO Bulletin, 2015*
18. Z. Obermeyer, B. Powers, **M. Makar**, N. Keating, D. Cutler, "Physician revealed preferences for hospice are strongly related to their patients' enrollment in hospice". *Health Affairs, 2015*
19. Z. Obermeyer, A. Clarke, **M. Makar**, J. Schuur, D. Cutler, "Emergency Care Use and the Medicare Hospice Benefit for Individuals with Cancer with a Poor Prognosis". *Journal of the American Geriatrics Society, 2016*
20. S. Skamene, I. Agarwal, **M. Makar**, et al., "Impact of a dedicated palliative radiation oncology service on the use of single fraction and hypofractionated radiation therapy among patients with bone metasteses". *Annals of Palliative Medicine, 2017; ASTRO abstract 2015*

Invited talks, podcasts and panels

1. Practical Big data workshop at the University of Michigan, 2022
2. NeuRIPS workshop on Algorithmic Fairness through the Lens of Causality and Robustness (Panelist), 2021
3. Trustworthy ML symposium initiative symposium (Panelist), 2021
4. Casual Inference podcast, 2021
5. Strategic reasoning group at the University of Michigan, 2021
6. Precision health seminar at the University of Michigan, 2021
7. e-Health and Artificial Intelligence, 2021

8. AI seminar at the University of Michigan, 2021
9. AI symposium at the University of Michigan, 2021
10. Center for Healthcare Engineering and Patient Safety at The University of Michigan, 2021
11. Causal Machine Learning group at Microsoft Research, 2021
12. Online causal inference seminar, 2021
13. Microsoft-MIT Trustworthy and Robust AI Collaboration, 2020
14. Microsoft-MIT Trustworthy and Robust AI Collaboration, 2019
15. AAAI Conference on Aritifical Intelligence, 2019.
16. Machine learning seminar, University of Massachusetts, 2018
17. AAAI Conference on Aritifical Intelligence, February 2018.
18. Computational Cardiovascular Research Group at MIT, 2017.
19. University of Michigan in Ann Arbor, July 2016
20. Harvard School of Public Health. October 2014

Professional Service

Teaching

Instructor, Machine learning, Big data science institute, University of Michigan	Summer 2022
Co-organizer & instructor, Whirlwind of Machine learning, MIT	IAP 2017
Teaching assistant, Machine learning for Healthcare, MIT	Spring 2017

Mentorship

Pengrun Huang, Undergraduate student	University of Michigan	Summer 2022
Jiayun Zheng, Masters of Engineering candidate	University of Michigan	2021-2022
Advaith Anand, Masters of Engineering candidate	MIT	2018-2019
Emily Mu, Masters of Engineering candidate	MIT	2018-2019
Agni Kumar, Masters of Engineering Candidate	MIT	2019-2020
Meghana Kamineni, Undergraduate researcher	MIT	2019-2021

Reviewing and other service

Co-organizer, 2022 ICML workshop on Spurious correlations, Invariance, and Stability
Reviewer, Causal inference workshop at UAI
Reviewer, UAI
Reviewer, ICML *Top 5% reviewer award*
Reviewer, NeuRIPS *Top 5% reviewer award*
Reviewer, Proceedings of the AAAI conference on AI
Reviewer, ICLR
Reviewer, JMLR
Reviewer, NeuRIPS workshop for Women in Machine Learning
Reviewer, NeuRIPS workshop on Machine Learning for health
Reviewer, Journal of Health Economics

Awards

NSF CRII (Sole PI)	2022-2024
e-HAIL summer student support	2022
Harvey Fellowship Award	2018- 2021
Ward McCarthy scholarship	2012
School of Behavioral and Social Sciences scholarship	2012
Economics Department Writing Award	2012
Life Members Scholarship	2012

Computer and Language Skills

Computing: Python, and R
Languages: English, Arabic, and French